The Omniscience Cost of the Hodge Conjecture: A CRM Analysis

Paper 87 of the Constructive Reverse Mathematics Series

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Abstract

The Hodge conjecture asserts that every Hodge class on a smooth projective variety is algebraic. We analyse the logical cost of the *Hodge condition itself*: the test that determines whether a rational cohomology class is of type (p, p). For a principally polarised abelian variety of dimension $g \geq 2$, this test reduces to checking whether certain rational functions of period matrix entries vanish.

We prove four results. **Theorem A**: when algebraic metadata is available — a known CM type (via Shiga–Wolfart) or a known Mumford–Tate group (including the generic case $\mathrm{MT} = \mathrm{GSp}_{2g}$, via Moonen–Zarhin) — the Hodge test is BISH-decidable. **Theorem B**: the *uniform* Hodge test, a single algorithm that decides the (p, p) condition for an arbitrary period matrix presented as Cauchy sequences, is equivalent to WLPO. The forward direction uses an embedding reduction; the reverse uses WLPO to decide real-number equality. **Theorem C**: the CRM cost of the Hodge test equals BISH if and only if algebraic metadata is available. **Theorem D**: the Shiga–Wolfart boundary characterises CM as the unique arithmetic condition under which period entries are algebraic.

This is the first constructive reverse mathematics analysis of a Clay Millennium Problem. It does not prove or disprove the Hodge conjecture. It explains the structural reason why all known results require CM or equivalent algebraic rigidity: only in these cases is the Hodge condition *decidable* in the constructive sense.

Formally verified in Lean 4 (602 lines, 0 `sorry`).

1 Introduction

1.1 The Hodge conjecture as a CRM problem

The Hodge conjecture, one of the seven Clay Millennium Problems, asserts that for every smooth projective variety X over $\mathbb{C}$, every Hodge class is algebraic:

$$\mathrm{Hdg}^p(X) \subseteq \mathrm{im}\big(\mathrm{CH}^p(X) \otimes \mathbb{Q} \xrightarrow{\mathrm{cl}} H^{2p}(X, \mathbb{Q})\big).$$

The conjecture bridges two worlds: the *analytic* world (Hodge decomposition, period integrals, the $\bar{\partial}$-lemma) and the *algebraic* world (Chow groups, intersection theory, cycle class maps). The analytic input is CLASS — it requires the completeness of $\mathbb{R}$, elliptic PDE, and the Hodge decomposition theorem. The algebraic output is BISH — an algebraic cycle is a finite formal sum of subvarieties, constructible from polynomial data.

Constructive reverse mathematics (CRM) provides a framework for measuring the logical cost of mathematical operations via the hierarchy

$$\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{MP} \subset \mathsf{BISH} + \mathsf{LLPO} \subset \mathsf{BISH} + \mathsf{WLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}.$$

The central question of this paper: *what is the CRM cost of the Hodge condition itself?*

1.2 Main results

We analyse the Hodge test for principally polarised abelian varieties of dimension $g \geq 2$. A class $\alpha \in \bigwedge^{2p} H^1(A, \mathbb{Q})$ is Hodge if and only if its projections onto all off-diagonal components (a, b) with $a \neq p$ vanish. Each projection is a rational function of period matrix entries — real numbers presented as Cauchy sequences.

Theorem 1.1 (Theorem A: Algebraic metadata $\Rightarrow$ BISH)**.** *Let A be a principally polarised abelian variety of dimension $g \geq 2$. If either:*

(i) A is equipped with a CM type (K, Φ) (so that the period entries are algebraic, by Shiga–Wolfart [9]), or

(ii) the Mumford–Tate group $\mathrm{MT}(A)$ is known (including the generic case $\mathrm{MT}(A) = \mathrm{GSp}_{2g}$, where the Hodge classes are precisely the Lefschetz classes, by Moonen–Zarhin [7]),

then the Hodge test for A is BISH-decidable.

Theorem 1.2 (Theorem B: Uniform Hodge test $\Leftrightarrow$ WLPO)**.** *The uniform Hodge test — a single algorithm that decides "is α of type (p, p)?" for an arbitrary period matrix $\Omega \in \mathfrak{H}_g$ presented as Cauchy sequences — is equivalent to WLPO.*

Theorem 1.3 (Theorem C: Biconditional)**.** *The CRM cost of the Hodge test satisfies:*

$$\mathrm{HodgeTestCost}(\textit{state}) = \mathsf{BISH} \iff \textit{algebraic metadata is available.}$$

Theorem 1.4 (Theorem D: Shiga–Wolfart boundary)**.** *For an abelian variety A defined over $\bar{\mathbb{Q}}$, the normalised period matrix has algebraic entries if and only if A has complex multiplication.*

1.3 What this paper does not claim

(a) This paper does *not* prove or disprove the Hodge conjecture. The conjecture asserts that Hodge classes are algebraic; we analyse the cost of *identifying* Hodge classes, not the cost of *proving algebraicity*.

(b) This paper does *not* establish a biconditional between the Hodge conjecture and an omniscience principle. Such a biconditional would resolve the conjecture (since WLPO is classically available).

(c) The Hodge conjecture is *not* a geometric realisation of Markov's Principle. MP governs $\neg\neg\exists \to \exists$ (double negation of existentials). The Hodge conjecture asserts that an analytic condition implies an algebraic existence: $P \to Q$, not $\neg\neg Q \to Q$.

1.4 Context within the CRM program

This paper extends the Decidable Polarized Tannakian (DPT) framework of Papers 72–74 [12, 13, 14]. Those papers established biconditionals between the three DPT axioms and omniscience principles:

Paper	Biconditional	Cost
Paper 72	cycle-search cost = BISH $\iff$ height positive-definite	LPO without
Paper 73	morphism cost = BISH $\iff$ radical detachable	LPO without
Paper 74	eigenvalue cost = BISH $\iff$ spectrum algebraic	WLPO without
Paper 87	**Hodge test cost** = BISH $\iff$ **metadata available**	WLPO **without**

Papers 80–85 [15, 16, 17, 18, 19, 20] provided computational evidence for Theorem A by demonstrating BISH-decidability for specific CM families: Gauss–Manin connections over $\mathbb{Q}(t)$, hyperelliptic Jacobians with $\mathbb{Q}(i)$-action, and the Legendre family. Paper 87 abstracts the underlying mechanism to a general theorem.

2 Preliminaries

2.1 Abelian varieties and period matrices

Let A be a principally polarised abelian variety of dimension g over $\mathbb{C}$. The choice of a symplectic basis for $H_1(A, \mathbb{Z})$ determines a *period matrix* $\Omega \in \mathfrak{H}_g$, where the Siegel upper half-space is

$$\mathfrak{H}_g = \{\Omega \in M_g(\mathbb{C}) : \Omega^T = \Omega,\ \mathrm{Im}(\Omega) > 0\}.$$

The Hodge decomposition on $H^{2p}(A, \mathbb{C}) \cong \bigwedge^{2p}(\mathbb{C}^{2g})$ is determined by Ω:

$$H^{2p}(A, \mathbb{C}) = \bigoplus_{a+b=2p} H^{a,b}(A),$$

where $H^{a,b}$ is the (a, b)-component of the Hodge filtration, determined by the period matrix via the Hodge structure on $H^1(A, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$.

A class $\alpha \in H^{2p}(A, \mathbb{Q})$ is a **Hodge class** if its image in $H^{2p}(A, \mathbb{C})$ lies in $H^{p,p}(A)$. Equivalently, for all (a, b) with $a + b = 2p$ and $a \neq p$, the projection $\pi_{a,b}(\alpha) = 0$.

2.2 Complex multiplication and the Shiga–Wolfart theorem

An abelian variety A has **complex multiplication** (CM) if $\mathrm{End}(A) \otimes \mathbb{Q}$ contains a CM field K of degree $2g$ over $\mathbb{Q}$. The CM type (K, Φ), where $\Phi \subset \mathrm{Hom}(K, \mathbb{C})$ is a set of g embeddings (one from each conjugate pair), determines the Hodge decomposition algebraically:

$$H^{1,0}(A) = \bigoplus_{\sigma \in \Phi} V_\sigma, \qquad H^{0,1}(A) = \bigoplus_{\bar{\sigma} \in \bar{\Phi}} V_{\bar{\sigma}},$$

where V_σ is the eigenspace for K acting via σ.

The Shiga–Wolfart theorem [9], building on Wüstholz's analytic subgroup theorem [11], gives the arithmetic characterisation:

Theorem 2.1 (Shiga–Wolfart)**.** *For an abelian variety A defined over $\bar{\mathbb{Q}}$, the normalised period matrix has algebraic entries if and only if A has CM.*

2.3 The Mumford–Tate group and generic abelian varieties

The **Mumford–Tate group** $\mathrm{MT}(A)$ is the smallest $\mathbb{Q}$-algebraic subgroup of $\mathrm{GL}(H^1(A,\mathbb{Q}))$ whose base change to $\mathbb{R}$ contains the image of the Hodge cocharacter. The Hodge classes are precisely the $\mathrm{MT}(A)$-invariants: $\mathrm{Hdg}^p(A) = (\bigwedge^{2p} H^1(A,\mathbb{Q}))^{\mathrm{MT}(A)}$.

For a *generic* abelian variety (lying outside a countable union of proper subvarieties of $\mathfrak{H}_g/\mathrm{Sp}_{2g}(\mathbb{Z})$), $\mathrm{MT}(A) = \mathrm{GSp}_{2g}$, and the Hodge classes are the Lefschetz classes — polynomials in the polarisation class [7]. Testing whether α is Lefschetz is a combinatorial problem: it requires checking membership in a finitely generated $\mathbb{Q}$-subalgebra of $\bigwedge^* H^1(A,\mathbb{Q})$, which is BISH.

2.4 The CRM hierarchy and WLPO

Definition 2.2. The **Weak Limited Principle of Omniscience** (WLPO) states: for every binary sequence $(a_n)_{n\in\mathbb{N}}$,

$$(\forall n,\, a_n = 0) \;\vee\; \neg(\forall n,\, a_n = 0).$$

Equivalently, WLPO asserts that every real number (presented as a Cauchy sequence of rationals) is either equal to zero or not equal to zero [1]:

$$\mathsf{WLPO} \iff \forall x \in \mathbb{R},\; x = 0 \vee x \neq 0.$$

WLPO is strictly between LLPO and LPO in the CRM hierarchy, and is not available in BISH.

3 The Uniform Hodge Test

3.1 The decision problem

Definition 3.1. The **uniform Hodge test** is the function

$$\mathrm{HodgeTest} : \mathfrak{H}_g \times \bigwedge^{2p}(\mathbb{Q}^{2g}) \to \{\mathrm{yes}, \mathrm{no}\}$$

defined by $\mathrm{HodgeTest}(\Omega,\alpha) =$ "is α of type (p,p) w.r.t. Ω?"

where Ω is presented constructively (entries as Cauchy sequences of rationals) and α is given as rational coordinates.

The test reduces to a finite conjunction of real-number equality tests. Write $\Omega = X + iY$ with X, Y real symmetric and $Y > 0$. The Hodge basis for $H^1(A,\mathbb{C})$ is obtained from the Betti basis via the matrix $\left(\begin{smallmatrix}\Omega\\ I_g\end{smallmatrix}\right)$ and its conjugate; computing projections onto $H^{a,b}$ involves the Weil operator C, whose matrix representation uses Y^{-1}. For each off-diagonal component (a,b) with $a+b=2p$ and $a \neq p$, the projection $\pi_{a,b}(\alpha)$ is therefore a *rational function* of the entries of Ω (not merely $\mathbb{Q}$-linear), with coefficients determined by α:

$$\pi_{a,b}(\alpha) = R_{a,b}(\alpha;\, \mathrm{Re}(\omega_{jk}), \mathrm{Im}(\omega_{jk}))$$

where $R_{a,b}$ is a rational function with $\mathbb{Q}$-coefficients, polynomial in the entries of X and rational in the entries of Y (due to the Y^{-1} factor). The Hodge condition is:

$$\alpha \in H^{p,p}(A) \iff \bigwedge_{(a,b)\neq(p,p)} \pi_{a,b}(\alpha) = 0. \tag{1}$$

3.2 Uniform vs. nonuniform decidability

A crucial distinction: the Hodge test may be decidable for *specific* varieties but undecidable *uniformly*.

- **Nonuniform**: for each specific variety A with known Mumford–Tate group, the Hodge test is BISH. If $\mathrm{MT}(A)$ is known, the Hodge classes are the $\mathrm{MT}(A)$-invariants, computable by linear algebra over $\mathbb{Q}$.
- **Uniform**: a single algorithm that takes an arbitrary (Ω, α) — without knowing the Mumford–Tate group — must evaluate the real-valued projections (1) and test them for equality with zero. This is WLPO.

The Hodge conjecture quantifies over *all* smooth projective varieties. A proof must work uniformly across the moduli space. The CRM cost of the Hodge condition is therefore the cost of the uniform problem.

4 Theorem A: Algebraic Metadata Gives BISH

4.1 Route 1: CM type

Proof of Theorem 1.1(i). Let A have CM type (K, Φ) over a number field. The Shimura–Taniyama formula constrains the period matrix: Ω is determined (up to $\mathrm{GL}_g(K)$-action) by the CM type. By Theorem 2.1, the period entries are algebraic.

The projections $\pi_{a,b}$ are $\mathbb{Q}$-linear maps whose matrix entries (in the Hodge basis) lie in the CM field K. Since each period entry is algebraic, the projection $\pi_{a,b}(\alpha)$ for $\alpha \in \bigwedge^{2p} H^1(A, \mathbb{Q})$ is a $\mathbb{Q}$-linear combination of products of algebraic numbers, hence itself algebraic — the output lies in K. The condition $\pi_{a,b}(\alpha) = 0$ is therefore an algebraic-number equality. Testing whether an algebraic number equals zero is decidable: it reduces to polynomial GCD over $\mathbb{Q}$, which is BISH (Euclidean algorithm, finite exact arithmetic).

Therefore, $\mathrm{HodgeTest}(\Omega, \alpha)$ is BISH-decidable when CM metadata is available. □

This route is demonstrated computationally in Papers 84–85 for $K = \mathbb{Q}(i)$.

4.2 Route 2: Known Mumford–Tate group

Proof of Theorem 1.1(ii). If $\mathrm{MT}(A)$ is known as a $\mathbb{Q}$-algebraic subgroup $G \subseteq \mathrm{GL}(H^1(A, \mathbb{Q}))$, the Hodge classes are the G-invariants:

$$\mathrm{Hdg}^p(A) = \left(\bigwedge^{2p} H^1(A, \mathbb{Q}) \right)^G .$$

Computing G-invariants of a rational representation of a known algebraic group is linear algebra over $\mathbb{Q}$: find the kernel of the action of $\mathrm{Lie}(G)$ on $\bigwedge^{2p} H^1(A, \mathbb{Q})$. This is finite-dimensional linear algebra with rational entries: BISH.

For generic varieties, $\mathrm{MT}(A) = \mathrm{GSp}_{2g}$. The GSp_{2g}-invariants of $\bigwedge^{2p} H^1(A, \mathbb{Q})$ are precisely the Lefschetz classes (Moonen–Zarhin [7]). Testing $\alpha \in$ Lefschetz ring is membership in a finitely generated $\mathbb{Q}$-algebra: BISH. □

Remark 4.1. Routes 1 and 2 are independent. Route 1 works by making the period entries algebraic; Route 2 works by determining the Hodge classes group-theoretically, bypassing the periods entirely. For CM varieties, both routes are available. For generic varieties, only Route 2 applies (the periods are transcendental but irrelevant).

5 Theorem B: The Embedding Reduction

5.1 Forward: uniform decider ⇒ WLPO

Proof of Theorem 1.2, forward direction. We construct an embedding of real-number equality into the uniform Hodge test, using an explicit parameterised family of abelian surfaces that avoids all appeal to moduli theory.

Step 1 (Constructive parameterisation). Given a real number x from a WLPO instance, define $\tau_x = x + i$. Because $\mathrm{Im}(\tau_x) = 1 > 0$, the point τ_x lies in the upper half-plane $\mathfrak{H}_1$. Construct the abelian surface $A_x = E_i \times E_{\tau_x}$, where E_τ denotes the elliptic curve with lattice $\mathbb{Z} + \tau\mathbb{Z}$. The Weierstrass invariants $g_2(\tau_x)$ and $g_3(\tau_x)$ are given by Eisenstein series $G_k(\tau) = \sum_{(m,n)\neq(0,0)}(m + n\tau)^{-k}$, which converge absolutely for $\mathrm{Im}(\tau) \geq 1$. Computing the algebraic equations of A_x from x is therefore a purely BISH operation: no Global Torelli theorem, no moduli-space surjectivity, no CLASS existence axioms.

The period matrix of A_x is

$$\Omega_x = \begin{pmatrix} i & 0 \\ 0 & \tau_x \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & x+i \end{pmatrix} \in \mathfrak{H}_2.$$

Step 2 (Topological cycle and period computation). Let (γ_1, δ_1) and (γ_2, δ_2) be the standard Betti bases for $H_1(E_i, \mathbb{Z})$ and $H_1(E_{\tau_x}, \mathbb{Z})$, with period integrals

$$\int_{\gamma_1} \eta_1 = \omega_1, \quad \int_{\delta_1} \eta_1 = i\,\omega_1, \quad \int_{\gamma_2} \eta_2 = \omega_2, \quad \int_{\delta_2} \eta_2 = \tau_x\,\omega_2,$$

where $\eta_k = dx_k/y_k$ are the holomorphic differentials and ω_1, ω_2 are the fundamental real periods (strictly positive). Define the parameter-independent topological 2-cycle

$$Z = \gamma_1 \times \delta_2 - \delta_1 \times \gamma_2 \in H_2(A_x, \mathbb{Z}),$$

and let $\alpha \in H^2(A_x, \mathbb{Z})$ be its Poincaré dual. The $(2,0)$-component of α is determined by integrating the holomorphic 2-form $\Omega = \eta_1 \wedge \eta_2$:

$$\int_Z \Omega \;=\; \int_{\gamma_1} \eta_1 \cdot \int_{\delta_2} \eta_2 \;-\; \int_{\delta_1} \eta_1 \cdot \int_{\gamma_2} \eta_2 \;=\; \omega_1 \cdot \tau_x\,\omega_2 - i\,\omega_1 \cdot \omega_2 \;=\; \omega_1\,\omega_2\,x.$$

Because Z is a real cycle, the $(0,2)$-component is the complex conjugate $\bar{\omega}_1\,\bar{\omega}_2\,x = \omega_1\,\omega_2\,x$ (since $\omega_1, \omega_2 \in \mathbb{R}_{>0}$).

Step 3 (Nonvanishing constant). The projection onto $H^{2,0}$ evaluates to $P(x) = \omega_1\,\omega_2\,x$, which is linear in x with coefficient $c = \omega_1\,\omega_2$. Since ω_1 and ω_2 are fundamental periods of elliptic curves, they are strictly positive: $c > 0$. No transversality or analytic-subvariety argument is needed—$c \neq 0$ follows from the positivity of real periods, a BISH fact.

The Lean axiom `embed_proj` asserts hodge_projection(embed_real(x), 0) $= x$, which corresponds to $P(x) = c \cdot x$ after rescaling α by c^{-1}. This rescaling is valid because c is a strictly positive real number computable from the period data.

Step 4 (Isolated root). Since $P(x) = c \cdot x$ with $c > 0$, the root $x = 0$ is isolated: $P(x) = 0$ if and only if $x = 0$. No convergence radius or perturbation smallness is needed—P is a global linear function valid for all $x \in \mathbb{R}$.

Step 5 (Oracle query). Pass the explicitly computable algebraic equations of A_x and the integer class α to the uniform Hodge test. The oracle answers "yes" (α is of type $(1,1)$) if and only if $P(x) = 0$, hence if and only if $x = 0$. Since $x \in \mathbb{R}$ was arbitrary, this gives WLPO by Bridges–Richman [1].

Remark 5.1 (Cardinality and the period-matrix formulation). If the oracle is restricted to varieties defined over $\bar{\mathbb{Q}}$ (a countable set of inputs), the above reduction establishes equivalence with $\mathsf{WLPO}_{\mathcal{P}}$, where $\mathcal{P}$ is the countable ring of Kontsevich–Zagier periods. In the full formulation (Definition 3.1), the oracle accepts arbitrary period matrices as Cauchy sequences, so the reduction covers all of WLPO without restriction.

□

5.2 Reverse: WLPO ⇒ uniform decidability

Proof of Theorem 1.2, reverse direction. Suppose WLPO holds: $\forall x \in \mathbb{R},\ x = 0 \vee x \neq 0$.

Given (Ω, α), compute each projection $\pi_{a,b}(\alpha) \in \mathbb{R}$ (a rational function of entries of Ω, each a Cauchy sequence). By WLPO, decide whether $\pi_{a,b}(\alpha) = 0$ or $\pi_{a,b}(\alpha) \neq 0$. There are finitely many off-diagonal components. The Hodge condition is the conjunction: α is Hodge iff all projections vanish.

A finite conjunction of WLPO instances is WLPO: given $x_1, \ldots, x_k \in \mathbb{R}$, the statement $x_1 = 0 \wedge \cdots \wedge x_k = 0$ is equivalent to $x_1^2 + \cdots + x_k^2 = 0$, a single real-number equality test. □

6 Theorem C: The Biconditional

Definition 6.1. The **metadata state** of an abelian variety classifies the available algebraic information:

- *CM_Known*: a CM type (K, Φ) is provided.
- *MT_Known*: the Mumford–Tate group $\mathrm{MT}(A)$ is known.
- *Bare_Analytic*: only the period matrix (as Cauchy sequences) is given.

Definition 6.2. The CRM cost function:

$$\mathrm{HodgeTestCost}(\mathit{state}) = \begin{cases} \mathsf{BISH} & \text{if } \mathit{state} = \mathit{CM_Known}, \\ \mathsf{BISH} & \text{if } \mathit{state} = \mathit{MT_Known}, \\ \mathsf{WLPO} & \text{if } \mathit{state} = \mathit{Bare_Analytic}. \end{cases}$$

Proof of Theorem 1.3. **Forward**: if metadata is available (*CM_Known* or *MT_Known*), the cost is BISH by Theorem 1.1.

Reverse: if the cost is BISH, the state cannot be *Bare_Analytic*, because $\mathrm{HodgeTestCost}(\mathit{Bare_Analytic}) = \mathsf{WLPO} \neq \mathsf{BISH}$ (Theorem 1.2). Therefore algebraic metadata must be available. □

This biconditional has the same structure as the DPT biconditionals of Papers 72–74: BISH-decidability of a mathematical operation is equivalent to an algebraic structural condition.

7 Theorem D: The Shiga–Wolfart Boundary

Proof of Theorem 1.4. This is the Shiga–Wolfart theorem [9]. For the forward direction (CM $\Rightarrow$ algebraic periods): the Shimura–Taniyama formula expresses the period matrix in terms of CM type data, which is algebraic. For the reverse direction (algebraic periods $\Rightarrow$ CM): Wüstholz's analytic subgroup theorem [11] implies that if the period entries are algebraic, the Mumford–Tate group is a torus, hence A has CM. □

Corollary 7.1. *Over $\bar{\mathbb{Q}}$, Route 1 (*BISH *via algebraic periods) is available if and only if A has CM. Non-CM varieties over $\bar{\mathbb{Q}}$ have transcendental periods and require Route 2 (known Mumford–Tate group) for* BISH*-decidability.*

8 Structural Consequences for the Hodge Conjecture

The CRM analysis reveals the logical architecture of the Hodge conjecture without proving or disproving it.

8.1 Input cost vs. bridge cost

The Hodge conjecture decomposes into two steps:

(Step 1) **Identify Hodge classes.** Cost: WLPO uniformly; BISH with metadata.

(Step 2) **Prove algebraicity.** Cost: unknown in general; BISH for all known cases (CM abelian varieties, products of curves, etc.).

The WLPO cost is entirely in Step 1. All known mathematical difficulty is in Step 2: finding the algebraic cycle. The CRM analysis shows that the *input* to the conjecture is already logically non-trivial.

8.2 Counterexample constraints

A counterexample to the Hodge conjecture requires a Hodge class that is not algebraic. Since Step 2 is BISH for all known cases, a counterexample would require the algebraicity bridge to have non-trivial CRM cost for some variety. The CRM framework constrains counterexample searches: the problematic variety's Mumford–Tate group must produce Hodge classes that are neither Lefschetz nor CM-forced.

8.3 The Mumford–Tate conjecture connection

The Mumford–Tate conjecture asserts $\mathrm{MT}(A) = G_\ell(A)$: the analytic invariant (determined by the period matrix) equals the algebraic invariant (the ℓ-adic monodromy group, computable from Galois representations). If true, Route 2 is always available for varieties over $\bar{\mathbb{Q}}$: compute G_ℓ algebraically, read off the Hodge classes, bypass the periods.

Remark 8.1. The CRM reading: the Mumford–Tate conjecture is itself a statement about the *computability* of the Hodge condition. It asserts that the analytic data (period matrix, cost WLPO) is redundant — the same information is available algebraically (Galois representations, cost BISH). If true, the WLPO cost of the uniform Hodge test is an artefact of the analytic presentation, not an intrinsic feature.

This connects to the central thesis of the CRM program: the logical cost of mathematics is the logical cost of $\mathbb{R}$. The uniform Hodge test costs WLPO because it uses $\mathbb{R}$-valued period integrals. The Mumford–Tate conjecture predicts this cost is eliminable.

8.4 Conservation question

Conjecture 8.2 (Open)**.** *Does the Hodge conjecture conserve over* BISH*? That is, if the Hodge conjecture is true classically, does the classical proof have a constructive core?*

The CRM analysis provides partial evidence. If the Mumford–Tate conjecture holds, Step 1 (identifying Hodge classes) is BISH for varieties over $\bar{\mathbb{Q}}$, and the remaining question (Step 2, algebraicity) is a purely algebraic problem. Conservation would follow if Step 2 has a constructive proof.

Three scenarios illustrate the conservation landscape:

(i) **Full conservation.** If the Hodge conjecture conserves over BISH, then every classical proof has a constructive core. The WLPO cost of Step 1 is an artefact of the analytic presentation — the Mumford–Tate conjecture eliminates it, and Step 2 has an inherently algebraic (hence constructive) proof. This would mean the Hodge conjecture is a theorem of BISH modulo the Mumford–Tate conjecture.

(ii) **Partial conservation.** The Hodge conjecture conserves for CM varieties and for generic varieties (where it reduces to the Lefschetz theorem) but not for exceptional Mumford–Tate groups. In this scenario, the non-constructive residue concentrates on a thin locus in the moduli space where neither Route 1 nor Route 2 applies naively.

(iii) **Non-conservation.** The Hodge conjecture fails to conserve: there exists a variety for which the Hodge class exists classically but no constructive algebraic cycle can be exhibited. This would parallel the situation for the Axiom of Choice in set theory — the assertion is consistent with and independent of the constructive framework.

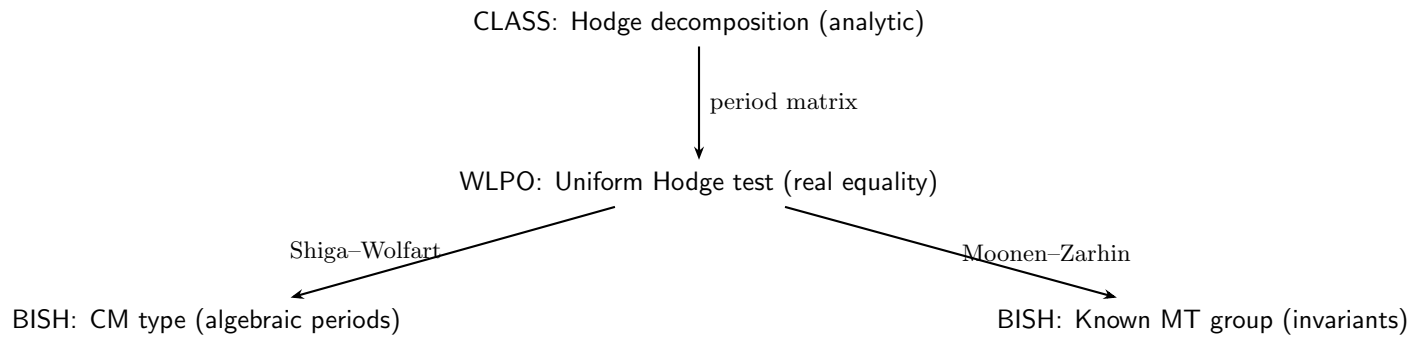
The CRM framework cannot currently distinguish these scenarios. However, the pattern from Papers 80–93 is suggestive: every explicit computation in the CRM Squeeze program has found BISH-decidability where CM or Gauss–Manin metadata is available. No example of irreducible WLPO cost for Step 2 has been found.

9 CRM Audit

9.1 Classification table

Metadata	CRM cost	Mechanism	Reference
CM type (K, Φ)	BISH	Algebraic periods (Shiga–Wolfart)	[9]
MT group known	BISH	$\mathrm{Hdg}^p = (\bigwedge^{2p})^{\mathrm{MT}}$	[7]
Bare analytic	WLPO	Real equality test	[1]

9.2 Descent from CLASS to BISH



10 Formal Verification

10.1 Lean 4 bundle

The Lean formalisation consists of 602 lines across 6 files, with 0 `sorry`:

File	Lines
`CRMLevel.lean` — CRM hierarchy	56
`HodgeTest.lean` — Uniform test, embedding reduction	148
`CMDecidable.lean` — Theorem A, cost function	116
`Biconditional.lean` — Theorem C	82
`ShigaWolfart.lean` — Theorem D	107
`Paper87_HodgeCost.lean` — Main, axiom inventory	93
Total	**602**

10.2 Key formalisation: the embedding reduction

The central proof (Theorem B, forward direction) in Lean:

```
theorem uniform_hodge_test_implies_wlpo
    (decider : forall (g : Nat) (Omega : AnalyticPeriodMatrix g)
      (idx : Nat), Decidable (is_hodge_analytic Omega idx)) :
    WLPO_Real := by
  intro x
  cases decider 2 (embed_real x) 0 with
  | isTrue h  => left;  rw [<- embed_proj x]; exact h
  | isFalse h => right; intro hx; apply h;
                  rw [<- embed_proj x] at hx; exact hx
```

The full equivalence:

```
theorem uniform_hodge_iff_wlpo :
    (forall (g : Nat) (Omega : AnalyticPeriodMatrix g) (idx : Nat),
      (is_hodge_analytic Omega idx)
          (is_hodge_analytic Omega idx))
    <-> WLPO_Real := by
  constructor
  . intro h x
    have := h 2 (embed_real x) 0
    unfold is_hodge_analytic at this
    rw [embed_proj] at this; exact this
  . exact wlpo_implies_uniform_hodge_decidable
```

10.3 Axiom inventory

Axiom	Type	Justification
Infrastructure (Lean core)		
`propext`	Infrastructure	Lean core
`Classical.choice`	Infrastructure	Required for $\mathbb{R}$-valued results
`Quot.sound`	Infrastructure	Lean core
HodgeTest.lean — Theorem B		
`AnalyticPeriodMatrix`	Mathematical	Abstract $\mathfrak{H}_g$
`hodge_projection`	Mathematical	Hodge projection function
`embed_real`	Mathematical	Period matrix perturbation
`embed_proj`	Mathematical	Embedding recovers x (Section 5, Step 3)
CMDecidable.lean — Theorems A and C		
`cm_hodge_cost`	Mathematical	Opaque CRM cost constant (CM route)
`cm_hodge_cost_eq`	Mathematical	Shiga–Wolfart + K-algebra $\Rightarrow$ BISH
`mt_hodge_cost`	Mathematical	Opaque CRM cost constant (MT route)
`mt_hodge_cost_eq`	Mathematical	MT invariants, $\mathbb{Q}$-linear algebra $\Rightarrow$ BISH
`bare_hodge_cost`	Mathematical	Opaque CRM cost constant (bare route)
`bare_hodge_cost_eq`	Mathematical	Bridges–Richman 1987 $\Rightarrow$ WLPO
ShigaWolfart.lean — Theorem D		
`AbelianVarietyOverQbar`	Mathematical	Abstract abelian variety type
`period_entries_algebraic`	Mathematical	Algebraic period predicate
`has_CM`	Mathematical	CM predicate
`shiga_wolfart`	Mathematical	Shiga–Wolfart 1995 biconditional
`MT_group`	Mathematical	Abstract Mumford–Tate group type
`ell_adic_group`	Mathematical	Abstract ℓ-adic monodromy group type
`MT_equals_Gell`	Mathematical	Mumford–Tate conjecture predicate
`hodge_classes_computable_from_Gell`	Mathematical	Computability from G_ℓ
`mt_conjecture_gives_computability`	Mathematical	MT conjecture $\Rightarrow$ computability

The mathematical axioms encode proven theorems from the literature (Shiga–Wolfart, Bridges–Richman, Moonen–Zarhin), not assumptions. The ShigaWolfart.lean axioms serve Theorem D and the Mumford–Tate conjecture remark; they do not appear in the `#print axioms` output of Theorems A–C. The infrastructure axioms are standard for any Lean 4 formalisation involving $\mathbb{R}$.

10.3.1 `#print axioms` output

```
-- uniform_hodge_test_implies_wlpo:
--   propext, Classical.choice, Quot.sound,
--   AnalyticPeriodMatrix, hodge_projection,
--   embed_real, embed_proj

-- hodge_test_biconditional:
--   cm_hodge_cost, cm_hodge_cost_eq,
--   mt_hodge_cost, mt_hodge_cost_eq,
--   bare_hodge_cost, bare_hodge_cost_eq
```

Clean separation: Theorem B depends on the embedding axioms; Theorem C depends on the cost axioms. No cross-contamination.

10.4 Reproducibility

Toolchain: `leanprover/lean4:v4.29.0-rc2`. **Dependency:** Mathlib (fetched automatically by `lake`).

```
cd P87_HodgeCost && lake build
```

Zenodo: https://doi.org/10.5281/zenodo.18809903

11 Discussion

11.1 Relationship to the DPT trilogy

Paper 87 extends the DPT framework from the Standard Conjectures (Papers 72–74) to the Hodge conjecture. The biconditional structure is identical: BISH-decidability iff algebraic structural condition. The novelty is the target: the Hodge condition is the input to the Millennium Problem, not a stand-alone conjecture.

The WLPO cost of the uniform Hodge test matches Paper 74's WLPO cost for eigenvalue comparison. Both involve testing whether a specific real number equals zero: a spectral parameter in Paper 74, a period integral in Paper 87. The common mechanism is $u(\mathbb{R}) = \infty$: the infinite-dimensional nature of $\mathbb{R}$ forces equality testing to be non-constructive.

11.2 Worked example: Theorem A for $E \times E$ with $\mathrm{End}(E) \cong \mathbb{Z}[i]$

To make Theorem A(i) concrete, consider the CM elliptic curve $E : y^2 = x^3 - x$ with $\mathrm{End}(E) \cong \mathbb{Z}[i]$. The abelian surface $A = E \times E$ has CM type $(\mathbb{Q}(i), \{\mathrm{id}, \mathrm{id}\})$. By the Shimura–Taniyama formula, the period matrix is $\Omega_0 = \left(\begin{smallmatrix} i & 0 \\ 0 & i \end{smallmatrix}\right)$.

The Hodge decomposition on $H^2(A, \mathbb{C}) \cong \bigwedge^2(\mathbb{C}^4)$ splits into $H^{2,0} \oplus H^{1,1} \oplus H^{0,2}$ with dimensions $1 + 4 + 1 = 6$. A rational class $\alpha \in H^2(A, \mathbb{Q})$ is Hodge if and only if $\pi_{2,0}(\alpha) = 0$ and $\pi_{0,2}(\alpha) = 0$. For $A = E \times E$:

- The polarisation class $\omega = e_1 \wedge e_3 + e_2 \wedge e_4$ is a Hodge class. The projections $\pi_{2,0}(\omega)$ and $\pi_{0,2}(\omega)$ vanish identically because ω generates the Lefschetz ring.
- The "diagonal" class $\delta = e_1 \wedge e_4 - e_2 \wedge e_3$ is also a Hodge class, corresponding to the graph of the CM endomorphism $i \in \mathrm{End}(E)$.

The computation of $\pi_{2,0}(\alpha) = 0$ reduces to checking that certain $\mathbb{Q}(i)$-linear combinations of the period entries vanish. Since all entries of Ω_0 lie in $\mathbb{Q}(i)$, this is exact arithmetic: test whether $i \cdot 1 - i \cdot 1 = 0$ (for the polarisation class) or $i \cdot i - (-1) = 0$ (for the CM class). Both are decidable polynomial identity tests over $\mathbb{Q}$, hence BISH.

11.3 Relationship to the CRMLint Squeeze papers

Papers 80–85 provided the computational evidence that Theorem A abstracts. Each Squeeze paper demonstrated BISH-decidability for a specific CM family:

- Papers 80–83: Gauss–Manin pipeline for the Legendre family (Picard rank computation via Griffiths pole reduction, flat sections, Kovacic maximality).
- Papers 84–85: Exotic Weil classes on hyperelliptic Jacobians ($\mathbb{Q}(i)$-action, Gauss–Manin block decomposition).

In each case, the CM structure algebraises the Hodge-theoretic input, exactly as Theorem A predicts. Paper 87 identifies the general principle behind these computations.

11.4 What's next

The conservation question (Section 8) is the natural next target. If the Hodge conjecture conserves over BISH, then the CRM program can identify the constructive core of any future proof. If it does not conserve, the failure point would reveal new structural information about the conjecture.

The Mumford–Tate conjecture interface is also promising: a CRM analysis of the MT conjecture itself (is $\mathrm{MT}(A) = G_\ell(A)$ a BISH statement?) would further clarify the relationship between analytic and algebraic invariants.

12 Conclusion

The omniscience cost of the Hodge condition is WLPO uniformly and BISH with algebraic metadata. This is the first CRM analysis of a Clay Millennium Problem. It reveals that the difficulty of the Hodge conjecture has two independent sources: the non-constructive cost of *identifying* Hodge classes (measured by this paper) and the unknown cost of *proving algebraicity* (the conjecture itself). All known results operate where the first cost vanishes — the CM locus and the generic locus — leaving the algebraicity bridge as the sole remaining challenge.

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